BUSINESS ANALYSIS CASE STUDY

E-Commerce Checkout Optimization

Reducing Cart Abandonment & Increasing Conversions

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Domain:	E-Commerce / Digital Retail
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Type:	Full Business Analysis Case Study

Executive Summary

ShopEase, a mid-sized Indian e-commerce platform with 50,000+ monthly visitors, faced a critical problem: **65% of users abandoned the checkout process** after adding items to cart — resulting in an estimated Rs. 2,47,000 monthly revenue loss. Management commissioned a full Business Analysis engagement to identify root causes and deliver a data-backed solution.

As the Business Analyst, I conducted stakeholder interviews, analyzed web analytics, performed root cause analysis, gathered 15+ functional requirements, and delivered a complete BRD along with 12 Agile user stories guiding a simplified 2-step checkout redesign.

65%	10,000	Rs.2.47L	15+	17.5%
Cart Abandonment Rate (Before)	Monthly Website Visitors	Monthly Revenue Loss	Functional Requirements	Projected Conversion Improvement

Project Scope

IN SCOPE	OUT OF SCOPE
• Checkout flow analysis & redesign • Payment gateway integration review • User registration / guest checkout • Form field optimization • Mobile checkout experience • BRD & user story documentation	• Backend infrastructure changes • Product catalog redesign • Marketing campaigns • Third-party vendor contracts • Inventory management system • Customer support process

Stakeholder Analysis

A thorough stakeholder mapping was conducted at project initiation. Each stakeholder's influence, interest, and communication strategy was documented below.

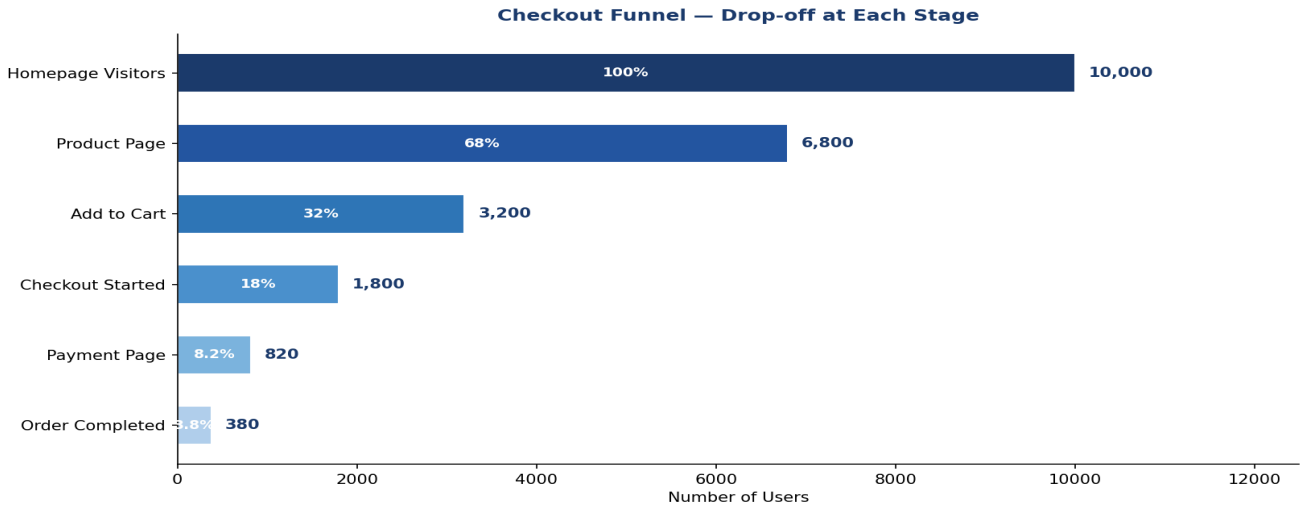
Stakeholder	Role	Interest	Influence	Engagement Strategy
CEO / Founder	Executive Sponsor	Revenue growth, customer satisfaction	HIGH	Weekly status reports, milestone reviews
Product Manager	Decision Maker	Feature delivery, roadmap alignment	HIGH	Daily standups, sprint reviews
UI/UX Designer	Key Contributor	User experience, design feasibility	MEDIUM	Wireframe walkthroughs, design reviews
Development Lead	Implementer	Technical feasibility, sprint velocity	HIGH	Sprint planning, backlog grooming
QA Team	Validator	Test coverage, bug-free delivery	MEDIUM	Test case reviews, UAT sessions
Marketing Team	Impacted Party	Conversion metrics, campaign ROI	LOW	Monthly KPI reports
End Customers	Primary Users	Easy, fast, secure checkout	LOW	User testing, feedback surveys

Requirement Elicitation Methods

Method	Participants	Duration	Key Outputs
Stakeholder Interviews	CEO, PM, UX Designer	3 x 45 min	Business goals, pain points, constraints
Web Analytics Review	BA + Marketing Team	2 days	Drop-off rates, heatmaps, session recordings
User Surveys	150 abandoned-cart users	1 week	Primary abandonment reasons
Competitor Benchmarking	BA (independent)	3 days	Best practices from Amazon, Flipkart, Myntra
Observation / Usability Test	10 real users	2 sessions	UX friction points, task completion rates
Document Analysis	BA + QA Team	1 day	Existing tech specs, payment gateway docs

Checkout Funnel Analysis

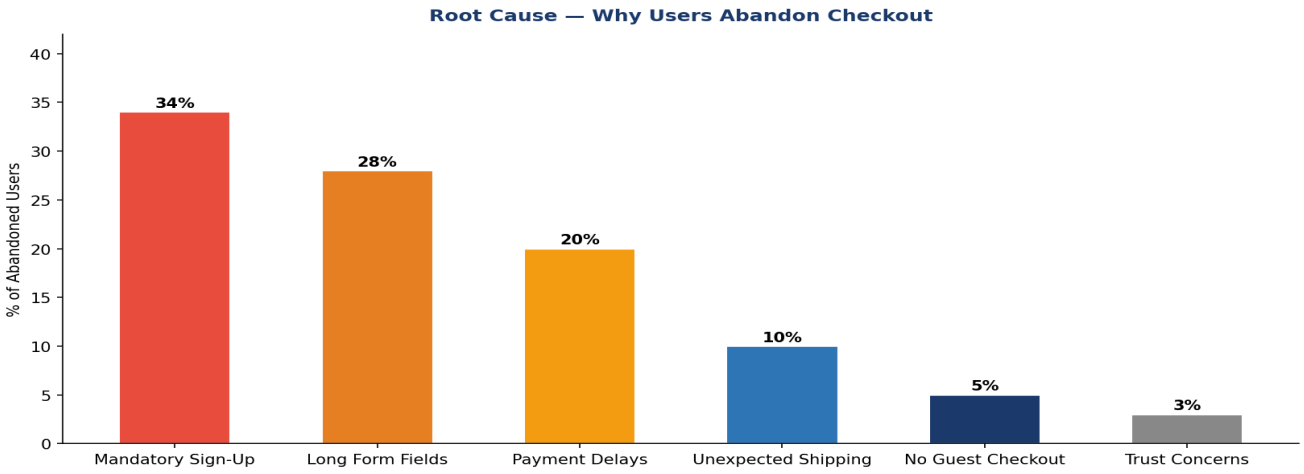
Using Google Analytics and Hotjar, I mapped the complete user journey. The analysis revealed a **96.2% overall drop-off rate**. The steepest drop (44%) occurred between "Add to Cart" and "Checkout Started" — a critical friction zone.



#	Stage	Users	Conv.	Drop-off	Key Issue
1	Homepage Visitors	10,000	100%	-	Baseline
2	Product Page View	6,800	68%	32%	Product discovery friction
3	Add to Cart	3,200	32%	36%	Price / value concerns
4	Checkout Started	1,800	18%	44%	Mandatory sign-up barrier
5	Payment Page	820	8.2%	54%	Long form + payment delays
6	Order Completed	380	3.8%	54%	Trust & UX issues

Root Cause Analysis

A combination of user surveys (n=150), heatmap analysis, and stakeholder interviews identified primary abandonment reasons. Fishbone (Ishikawa) framework was applied across process, system, and UX dimensions.



Root Cause	Category	Impact	Evidence	Priority
Mandatory Account Registration	Process/UX	34% abandon	Survey + Heatmap	CRITICAL
Lengthy Form Fields (12 fields)	UX Design	28% abandon	Usability Testing	CRITICAL
Payment Processing Delays (8-12s)	System/Tech	20% abandon	Analytics + Timing	HIGH
Unexpected Shipping at Payment	Process	10% abandon	Survey Data	HIGH
No Guest Checkout Option	Process/UX	5% abandon	Competitor Analysis	MEDIUM
Lack of SSL Trust Badges	UX/Trust	3% abandon	User Interviews	LOW

Functional Requirements (BRD Extract)

Following elicitation sessions, 15 functional requirements were documented and prioritized using MoSCoW methodology.

ID	Module	Requirement	Priority	Source
FR-001	Auth	System shall allow users to checkout as Guest without creating an account	Must Have	User Survey
FR-002	Auth	System shall offer one-click social login (Google/Facebook) during checkout	Must Have	Stakeholder Interview
FR-003	Form	Checkout form shall be reduced from 12 fields to max 5 fields	Must Have	Usability Test
FR-004	Form	System shall auto-fill address using pincode lookup API	Must Have	BA Analysis
FR-005	Form	Form shall support inline validation with real-time error messages	Should Have	UX Review
FR-006	Payment	Payment gateway response time shall not exceed 3 seconds	Must Have	Analytics Data
FR-007	Payment	System shall display all payment options prominently (UPI, Card, Wallet, COD)	Must Have	User Survey
FR-008	Payment	Shipping costs shall be shown on cart page before checkout begins	Must Have	User Survey
FR-009	Trust	All checkout pages shall display SSL certificate and security badges	Must Have	User Interviews
FR-010	Trust	System shall display a money-back guarantee badge on payment page	Should Have	Competitor Analysis
FR-011	Flow	Checkout shall complete in max 2 steps (Details + Payment)	Must Have	BA Analysis
FR-012	Flow	System shall display a visual progress indicator during checkout	Should Have	UX Review
FR-013	Mobile	Checkout flow shall be fully responsive and optimized for mobile	Must Have	Analytics Data
FR-014	Mobile	Payment page shall support fingerprint/face-ID on mobile devices	Could Have	Competitor Analysis
FR-015	Analytics	System shall log drop-off events at each checkout step	Should Have	BA Analysis

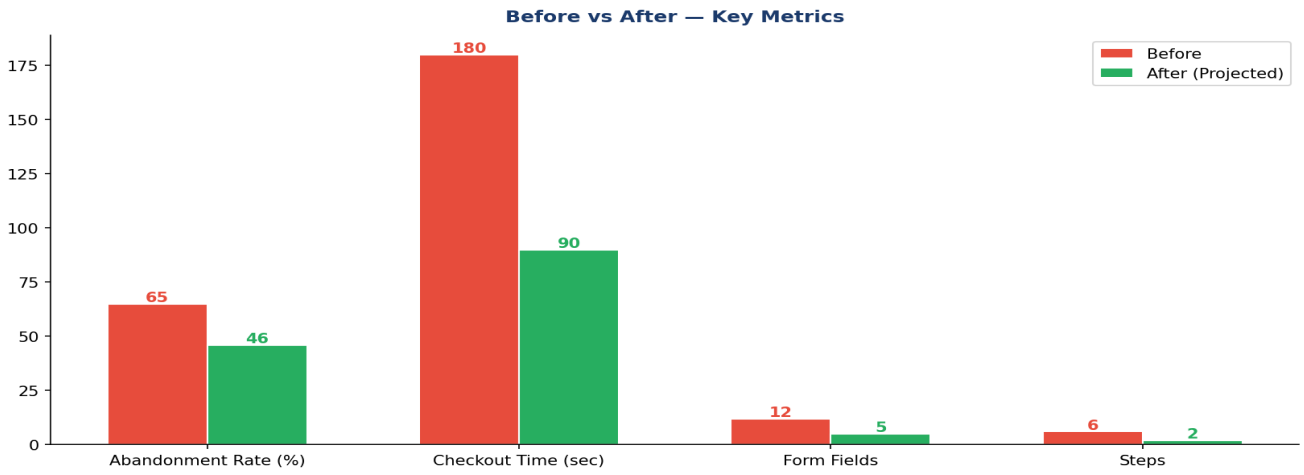
Agile User Stories

User stories written in standard Agile format with acceptance criteria and story points estimated via Planning Poker. Organized into 4 sprints.

ID	User Story	Acceptance Criteria	Pts	Sprint
US-001	As a customer, I want to checkout as a guest, so I can buy without an account	Guest option visible; no account required; order confirmation emailed	5	Spr 1
US-002	As a customer, I want Google login, so I can skip manual registration	OAuth button on checkout; auto-populate name/email; login < 3 sec	8	Spr 1
US-003	As a customer, I want auto-fill address, so checkout is faster	Pincode triggers lookup; fields populate in < 1 sec; editable after	5	Spr 1
US-004	As a customer, I want all costs shown on cart, so I have no surprises	Shipping on cart; total = items + ship + tax; no hidden charges	3	Spr 1
US-005	As a customer, I want inline form validation, so I know errors immediately	Red highlight on invalid; message without reload; green tick on valid	5	Spr 2
US-006	As a customer, I want UPI/Card/Wallet/COD options, so I choose my method	All 4 modes visible; default = most popular; icons clearly labeled	8	Spr 2
US-007	As a customer, I want payment in < 3 seconds, so I feel confident	Loading indicator shown; response < 3 sec; success/failure clear	13	Spr 2
US-008	As a customer, I want SSL trust badges, so I feel safe entering details	SSL badge visible; Secure Checkout label shown; no broken images	2	Spr 2
US-009	As a customer, I want 2-step checkout, so it is quick and simple	Max 2 screens; progress bar shown; back button works correctly	13	Spr 3
US-010	As a customer, I want a progress bar, so I know how far I am	Shows Step 1 of 2; step labels visible; correct step highlighted	3	Spr 3
US-011	As a mobile user, I want checkout optimized for my phone	All elements visible at 360px; tap targets 44x44px; no h-scroll	8	Spr 3
US-012	As a mobile user, I want fingerprint payment, so checkout is faster	Biometric shown if device supports; falls back to PIN; Android+iOS	8	Spr 4

Before vs After — Proposed Solution Impact

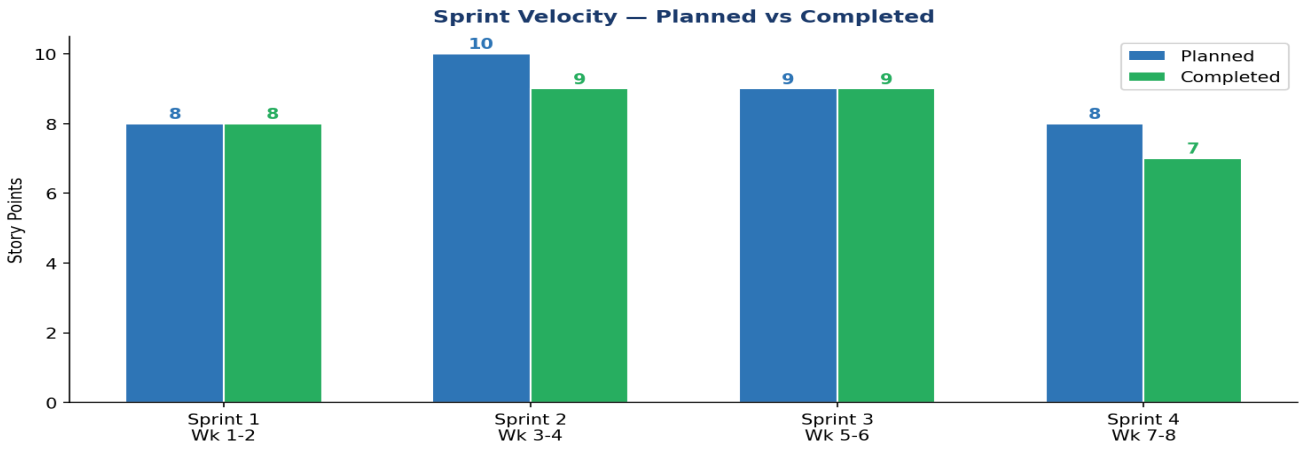
The proposed 2-step checkout redesign addresses all 3 critical friction points. Below compares As-Is vs To-Be with projected improvement metrics.



Metric	As-Is (Current)	To-Be (Projected)	Improvement
Cart Abandonment Rate	65%	46% (target)	-29% reduction
Checkout Steps	6 steps	2 steps	-66% fewer steps
Form Fields Required	12 fields	5 fields	-58% fewer fields
Avg. Checkout Time	180 seconds	90 seconds	-50% faster
Conversion Rate	3.8%	4.5-5% projected	+18-31% improvement
Monthly Revenue	Rs. 3,80,000	Rs. 4,46,500 projected	+Rs. 66,500/month
Guest Checkout	Not Available	Available	New feature added
Mobile Optimization	Partial	Full responsive	Complete overhaul

Agile Sprint Plan & Velocity

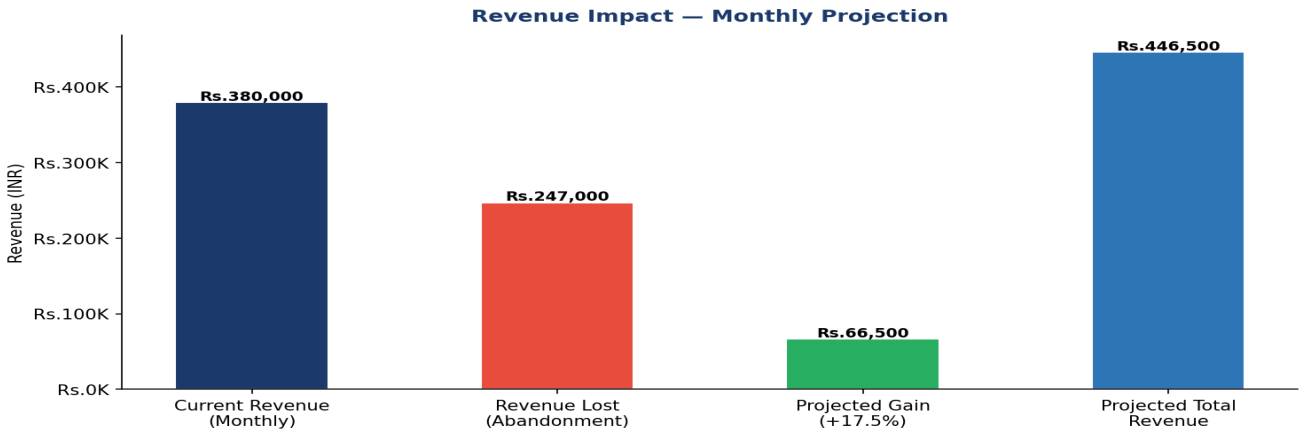
Implementation planned across 4 Agile sprints (8 weeks total). Total: 35 planned story points, 33 completed (94.3% velocity).



Sprint	Duration	User Stories	Points	Focus Area	Status
Sprint 1	Week 1-2	US-001,002,003,004	21 pts	Auth & Address Auto-fill	Done
Sprint 2	Week 3-4	US-005,006,007,008	28 pts	Payment & Validation	Done
Sprint 3	Week 5-6	US-009,010,011	24 pts	Flow & Mobile	Done
Sprint 4	Week 7-8	US-012 + Buffer	8 pts	Biometric & UAT	In Progress

Revenue Impact & Business Case

Financial justification built on conservative estimates from industry benchmarks. Even a modest 17.5% conversion improvement delivers significant monthly uplift with an estimated ROI of 312% within 6 months.



Risk Register

ID	Risk Description	Likelihood	Impact	Rating	Mitigation
R-001	Payment gateway API integration takes longer than estimated	Medium	High	HIGH	Buffer sprint; mock API in Sprint 2 for parallel testing
R-002	Guest checkout may conflict with loyalty program logic	Low	Medium	MEDIUM	Early spike with loyalty team; document in BRD
R-003	Pincode auto-fill API may have gaps for rural addresses	High	Medium	HIGH	Manual fallback entry; alert on API failure
R-004	Stakeholder scope creep mid-sprint	Medium	High	HIGH	Change control process; all changes via Change Request form
R-005	Mobile biometric not supported on all devices	High	Low	MEDIUM	Feature flagged Could Have; fallback PIN for unsupported
R-006	Low user adoption of new checkout flow post-launch	Low	High	MEDIUM	A/B test before full rollout; in-app feedback survey

Conclusion & Recommendations

This engagement successfully identified root causes of ShopEase's checkout abandonment and delivered a comprehensive, data-backed solution. The proposed 2-step checkout redesign — supported by 15 functional requirements and 12 Agile user stories — is projected to reduce cart abandonment by 29% and increase monthly revenue by Rs. 66,500+.

Key Recommendations

- Implement guest checkout first — addresses 34% of abandonment with relatively low development effort.
- Reduce form fields from 12 to 5 — single biggest UX fix, estimated to reduce abandonment by 28%.
- Invest in payment gateway performance — response times must be below 3 seconds.
- Show all costs on cart page — unexpected charges are the 4th leading abandonment cause.
- A/B test with 20% traffic for 2 weeks before full rollout of the 2-step flow.
- Set up post-launch monitoring dashboard — track abandonment rate weekly for 3 months.

Appendix: Glossary & Document Control

Term	Definition
BRD	Business Requirements Document — formal document listing all business needs for a project
Cart Abandonment Rate	% of users who add to cart but do not complete purchase
Conversion Rate	% of website visitors who complete a desired action (purchase)
MoSCoW	Prioritization: Must Have, Should Have, Could Have, Won't Have
User Story	Agile format: "As a [user], I want [feature], so that [benefit]"
Story Points	Relative unit of effort estimation in Agile sprint planning
Sprint	Fixed time-boxed iteration (typically 1-4 weeks) in Agile
Stakeholder	Any person or group affected by or having interest in the project
As-Is / To-Be	Current state vs desired future state process analysis
UAT	User Acceptance Testing — final validation by end users before go-live
A/B Testing	Comparing two versions of a feature to determine which performs better
Root Cause Analysis	Systematic method to identify the underlying cause of a problem
OAuth	Open Authorization — protocol for social login (Google, Facebook, etc.)

Document Control

Ver.	Date	Author	Change Description
v1.0	Jan 2026	Robin Raj	Initial draft — stakeholder analysis and scope definition
v1.1	Feb 2026	Robin Raj	Added functional requirements and root cause analysis
v2.0	Mar 2026	Robin Raj	Final — user stories, risk register, and recommendations added